Linear Algebra 2026 Homework 7

Due: November 13, 2026

Problems should be solved in English and calculation steps are required. For all T/F questions, you don't need to justify your answer.

All notations can be found in the book "Linear Algebra — the fourth edition" by Stephen H. Friedberg, Arnold J. Insel, Lawrence E. Spence. If you cannot locate them, feel free to contact the TAs.

Problem 1 [3 pts]. Label the following statements as true or false.

(a) The function $\det : \mathsf{M}_{2\times 2}(F) \to F$ is a linear transformation.

(b) The determinant of a 2×2 matrix is a linear function of each row of the matrix when the other row is held fixed.

(c) If $A \in \mathsf{M}_{2\times 2}(F)$ and $\det(A) = 0$, then A is invertible.

Problem 2 [5 pts]. Compute the determinant of the following matrix in $\mathsf{M}_{2\times 2}(R)$.

$$\begin{pmatrix} 6 & -3 \\ 2 & 4 \end{pmatrix}$$

Problem 3 [5 pts]. Compute the determinant of the following matrix in $\mathsf{M}_{2\times 2}(C)$.

$$\begin{pmatrix} -1+i & 1-4i \\ 3+2i & 2-3i \end{pmatrix}$$

Problem 4 [5 pts]. Prove that $\det(AB) = \det(A) \cdot \det(B)$ for any $A, B \in \mathsf{M}_{2\times 2}(F)$.

Problem 5 [8 pts]. Label the following statements as true or false.

(a) The function $\det : \mathsf{M}_{n\times n}(F) \to F$ is a linear transformation.

(b) The determinant of a square matrix can be evaluated by cofactor expansion along any row.

(c) If two rows of a square matrix A are identical, then $\det(A) = 0$.

(d) If B is a matrix obtained from a square matrix A by interchanging any two rows, then $\det(B) = -\det(A)$.

(e) If B is a matrix obtained from a square matrix A by multiplying a row of A by a scalar, then $\det(B) = \det(A)$.

(f) If B is a matrix obtained from a square matrix A by adding k times row i to row j, then $\det(B) = k\det(A)$.

(g) If $A \in \mathsf{M}_{n\times n}(F)$ has rank n, then $\det(A) = 0$.

(h) The determinant of an upper triangular matrix equals the product of its diagonal entries.

Problem 6 [10 pts]. Evaluate the determinant of the given matrix by cofactor expansion along the first row.

$$\begin{pmatrix} 0 & 1 & 2 \\ -1 & 0 & -3 \\ 2 & 3 & 0 \end{pmatrix}$$

Problem 7 [5 pts]. Evaluate the determinant of the given matrix by cofactor expansion along the third row.

$$\begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 5 \\ -1 & 3 & 0 \end{pmatrix}$$

Problem 8 [10 pts]. Prove that $\det(kA) = k^n \det(A)$ for any $A \in \mathsf{M}_{n\times n}(F)$.

Problem 9 [10 pts]. Suppose that $M \in \mathsf{M}_{n\times n}(F)$ can be written in the form

$$M = \begin{pmatrix} A & B \\ O & I \end{pmatrix},$$

where A is a square matrix. Prove that $\det(M) = \det(A)$.

Problem 10 [10 pts]. Prove that if $M \in \mathsf{M}_{n\times n}(F)$ can be written in the form

$$M = \begin{pmatrix} A & B \\ O & C \end{pmatrix},$$

where A and C are square matrices, then $\det(M) = \det(A) \cdot \det(C)$.

Problem 11 [20 pts]. Let L be a 4×4 unit lower triangular matrix and let U be a 4×5 row echelon matrix. For each of the following matrices, if A can be LU decomposed, find L, U such that $A = LU$, otherwise, find a permutation matrix P and L, U such that $PA = LU$.

(a)

$$A_1 = \begin{bmatrix} 2 & 1 & 0 & 1 & 3 \\ 4 & 3 & 1 & 2 & 5 \\ 6 & 5 & 3 & 4 & 7 \\ 8 & 7 & 6 & 5 & 9 \end{bmatrix}.$$

(b)

$$A_2 = \begin{bmatrix} 1 & 2 & 0 & 1 & 3 \\ 3 & 6 & 1 & 4 & 10 \\ 2 & 5 & 1 & 2 & 8 \\ 4 & 9 & 3 & 7 & 16 \end{bmatrix}.$$

Problem 12 [15 pts]. Let

$$A = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 2 & 1 & 0 & 0 \end{pmatrix}, \quad b_1 = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \quad b_2 = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}, \quad b_3 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

Solve the systems of linear equations $Ax = b_1$, $Ax = b_2$, and $Ax = b_3$ using LU decomposition. (If you use other methods to calculate, you will get zero points.)